Fuel Energy: Operation of an Internal Combustion Engine

Objective of Experiment No. 1

To verify the operating principle of a gasoline internal combustion engine.

Materials

- Perforated plexiglass cylinder with sponge stopper
- Spray bottle (with 10 ml of denatured alcohol – not included)
- Gas lighter

Procedure

STEP 1

Place the plexiglass cylinder on the work surface in a nearly horizontal position. Using the spray bottle, nebulize 2–3 sprays of alcohol to distribute it evenly on the inner surface of the tube.

STEP 2

Close the open end of the cylinder with the sponge stopper.

STEP 3

Insert the tip of the gas lighter into the perforated base of the cylinder, then ignite it to generate sparks and trigger combustion.

Experimental Questions

- What happens if the amount of alcohol sprayed inside the tube increases?
- Does a greater amount of fuel ignite effectively under conditions of limited oxidizer (oxygen in the air)?
- What happens if, instead of denatured alcohol, a distillate with an alcohol content lower than 30% is sprayed?
- Is the alcohol concentration in the fuel directly proportional to its calorific value?

- Using a syringe filled with air, repeat the experiment by introducing additional air into the cylinder before ignition (spark from the lighter).
  With the same amount of alcohol sprayed, what happens to the combustion flame? Will it be more or less intense compared to the normal condition with less air?
  (principle of supercharging)